

DermAI: Multimodal Skin Disease Detection & Consultation Platform

Comprehensive architectural reference, deep learning training pipeline, explainability engine, and complete collection of viva / presentation questions for examiners.

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Project Type: B.Tech CSE
Major Project

Datasets: HAM10000 +
ISIC 2019

Stack: React 18 + Node.js Express + Python FastAPI +
PyTorch/TF + Gemini

1 End-to-End System Working (Poora System Kaise Kaam Karta Hai?)

DermAI ek **Multimodal Microservices Architecture** par bana hai. Jab ek patient ya user website kholta hai, toh yeh complete sequence execute hoti hai:

STEP 1: CLIENT TIER (React 18 + Vite + Tailwind CSS) [Port 5173] |

- User patient photo upload karta hai (Drag & Drop) + metadata deta hai (Age, Gender) |
- JWT Auth token ke sath request API Gateway ko bheji jaati hai |

HTTP POST /api/scans (Multipart/Form-Data)

STEP 2: API GATEWAY & BUSINESS CORE (Node.js + Express.js) [Port 5000] |

- JWT Token verify karta hai, file sanitize karta hai (10MB size limit) |
- Image ko local disk par store karta hai aur Python ML Service ko forward karta hai |

Forward Image to FastAPI /predict

STEP 3: DEEP LEARNING INFERENCE ENGINE (Python FastAPI) [Port 8000] |

- **CNN Branch (EfficientNet):** Image (224x224) se visual textures aur borders nikalta hai |
- **RNN Branch (Bi-GRU):** Patient age, sex, body location sequence ko process karta hai |
- **Fusion Layer:** Dono vectors Concatenate hokar 7 disease probabilities banate hain |
- **Grad-CAM:** Last conv layer ke gradients se 2D color visual heatmap banata hai |

Returns JSON: Top Disease + Confidence + Heatmap

STEP 4: CLINICAL SAFETY & BILINGUAL AI CONSULTATION (Google Gemini + Safety RAG) |

- Clinical Safety Filter: Dangerous prescription drugs (oral steroids) ko block karega |
- Gemini LLM: English default, Hindi/Hinglish mein puche toh Hindi mein guidance dega |

Coordinates / City query

STEP 5: GEOLOCATION DOCTOR FINDER (OpenStreetMap Nominatim + Overpass API) |

- Patient ke GPS location ke 5-15km radius mein verified dermatologists dhoondhta hai |
- Consultation fee, doctor profile, clinic address aur 1-click Google Maps link deta hai |

2 Deep Learning Model Architecture (CNN + RNN Multimodal Code)

Model definition code `ml-service/train.py` ke function `build_model()` mein hai:

```
def build_model(num_classes):
    # — BRANCH 1: CNN (Image Processing) —
    image_input = keras.Input(shape=(224, 224, 3), name="image")
    base = keras.applications.EfficientNetB0(include_top=False, weights="imagenet", input_tensor=image_input)
    base.trainable = False # Pre-trained ImageNet weights freeze kiye
    x = layers.GlobalAveragePooling2D()(base.output)
    x = layers.Dropout(0.25)(x)

    # — BRANCH 2: RNN (Clinical Metadata Processing) —
    text_input = keras.Input(shape=(64,), dtype="int32", name="context_tokens")
    t = layers.Embedding(input_dim=10000, output_dim=64, mask_zero=True)(text_input)
    t = layers.Bidirectional(layers.GRU(64))(t) # Bidirectional Gated Recurrent Unit
    t = layers.Dropout(0.25)(t)

    # — BRANCH 3: MULTIMODAL FUSION & CLASSIFIER —
    fused = layers.Concatenate()([x, t]) # CNN image vector + RNN text vector fused!
    fused = layers.Dense(256, activation="relu")(fused)
    fused = layers.Dropout(0.35)(fused)
    output = layers.Dense(num_classes, activation="softmax")(fused)

    model = keras.Model(inputs=[image_input, text_input], outputs=output)
    model.compile(optimizer=keras.optimizers.Adam(1e-3), loss="sparse_categorical_crossentropy", metrics=
["accuracy"])
    return model
```

CNN Branch (Image)

Backbone: EfficientNetB0 (Compound scaling of depth, width, resolution).
Lesion ke colors, border asymmetry aur pigments extract karta hai.

RNN Branch (Text)

Backbone: Bidirectional GRU with Embedding. Patient age bucket, sex aur body location ko text sequence ke taur par aage-peeche dono side se learn karta hai.

Multimodal Fusion

Technique: Feature Concatenation. Dense(256) ReLU layer non-linearity deti hai aur Softmax layer 7 disease classes ki probabilities deti hai.

3 Datasets & The 7 Clinical Diagnostic Classes

Class Code	Disease Full Name	Clinical Features & Symptoms	Risk Severity
mel	Melanoma	Malignant skin cancer of melanocytes. Fast metastasis, ABCDE criteria (Asymmetry, Border, Color, Diameter >6mm, Evolving).	HIGH RISK
bcc	Basal Cell Carcinoma	Most common skin malignancy; pearly, waxy nodule with telangiectasia from long-term UV sun exposure.	MODERATE-HIGH
akiec	Actinic Keratoses	Precancerous crusty, scaly sun patch. Can turn into Squamous Cell Carcinoma (SCC) if ignored.	PRECANCEROUS
bkl	Benign Keratosis	Seborrheic keratosis / solar lentigo; harmless warty, stuck-on skin appearance in aging population.	BENIGN / LOW
nv	Melanocytic Nevus	Standard benign mole, birthmark. Symmetrical borders and uniform pigmentation.	BENIGN / NORMAL
vasc	Vascular Lesion	Cherry angiomas, pyogenic granulomas, vascular malformations formed by dilated blood capillaries.	BENIGN / LOW

Class Code	Disease Full Name	Clinical Features & Symptoms	Risk Severity
df	Dermatofibroma	Harmless fibrous skin nodule developing after minor trauma, insect bite, or ingrown hair.	BENIGN / LOW

4 Top 25 Viva Questions & Answers (Teachers Ke Sawaal)

Teacher ya external examiner presentation ke dauran yahi questions puchte hain. Inhe bilkul rat lein:

Category A: Deep Learning & Neural Network Fundamentals

Q1 Activation functions kaunse use huye hain aur kyu?

Answer: Hamare model mein do main activation functions hain: (1) **ReLU** hidden Dense(256) layer mein use hua hai non-linearity introduce karne aur vanishing gradients prevent karne ke liye. (2) **Softmax** final output layer mein use hua hai taaki 7 skin disease classes ke raw logits ko 0 se 1 ke probability distribution (sum = 100%) mein convert kiya ja sake.

Q2 Loss function kaunsa hai aur kyu? `sparse\_categorical\_crossentropy` vs `categorical\_crossentropy`?

Answer: Humne `sparse_categorical_crossentropy` use kiya hai kyunki hamare ground truth disease labels integer format mein hain (0, 1, 2, ... 6). Agar labels One-Hot Encoded vectors `[0, 1, 0, 0, 0, 0, 0]` hote toh hum `categorical_crossentropy` use karte. Dono ka math same hai, lekin sparse version memory save karta hai.

Q3 Optimizer kaunsa use kiya hai aur learning rate kya hai?

Answer: Adam Optimizer (Adaptive Moment Estimation) with a learning rate of `1e-3` (0.001). Adam momentum aur RMSprop dono ke concepts ko combine karta hai, jisse sparse gradients par bhi weights quickly aur smoothly converge hote hain.

Q4 Overfitting ko rokne ke liye model mein kya techniques lagayi hain?

Answer: Humne 4 levels of regularization lagayi hai: (1) **Dropout layers** (0.25 after pooling, 0.35 before dense layer) jo training ke time randomly neurons deactivate karti hain. (2) **Data Augmentation** (Random flip, rotation, contrast). (3) **Transfer Learning** (Pretrained weights freeze karke). (4) **Stratified Train/Val split** taaki validation set training distribution ko accurately represent kare.

Q5 GlobalAveragePooling2D aur Flatten layer mein kya difference hai? Humne GAP kyu use kiya?

Answer: `Flatten` layer saare 2D feature maps ko linear array bana deti hai jisse millions of trainable parameters ban jate hain aur overfitting ka risk badh jata hai. **GlobalAveragePooling2D** har feature map ka single average value nikal kar parameters ko drastic reduce karta hai aur spatial translations ke against model ko invariant banata hai.

Category B: CNN vs RNN Multimodal Architecture

Q6 Aapka model Multimodal kyu hai? Sirf CNN se kaam kyu nahi chalaya?

Answer: Real clinical dermatology mein ek certified doctor sirf photo dekh kar diagnose nahi karta. Patient ki age (e.g. skin cancer elderly mein zyada hota hai), gender, aur lesion ki body location (face, back, palm) diagnostic accuracy ke liye extremely critical hoti hai. Multimodal architecture visual signals (CNN) aur clinical metadata (RNN) dono ko fuse karke diagnosis banata hai, bilkul real doctor ki tarah.

Q7 EfficientNetB0 ko hi CNN backbone kyu chuna? ResNet ya VGG kyu nahi?

Answer: VGG-16/19 bahut heavy hain (138M parameters) aur slow hain. **EfficientNetB0 (Compound Scaling)** network depth, width, aur image resolution ko scientifically balance karta hai. Yeh sirf ~5.3 Million parameters mein ResNet-50 se better accuracy deta hai, jisse hamara API response time under 100ms rehta hai.

Q8 RNN branch mein GRU kyu use kiya, SimpleRNN ya LSTM kyu nahi?

Answer: Standard SimpleRNN mein short memory aur vanishing gradient problem hoti hai. LSTM mein 3 gates (Input, Forget, Output) hote hain jo heavy hain. **GRU (Gated Recurrent Unit)** sirf 2 gates (Reset aur Update) use karta hai, isliye LSTM jitni expressive power ke sath faster train hota hai aur short clinical sentences ke liye optimal hai.

Q9 Bidirectional RNN ka kya benefit hua yahan?

Answer: Normal RNN text ko sirf left-to-right padhta hai. **Bidirectional GRU** text sequence ko forward aur backward dono directions mein padhta hai. Isse agar sentence hai "back male age_40s", toh context tokens ka aapsi correlation dono taraf se learn hota hai.

Category C: Datasets, Splitting & Explainable AI (Grad-CAM)

Q10 Kaunse datasets use huye hain? Dono ka description do.

Answer: Humne 2 global benchmark datasets use kiye hain: (1) **HAM10000:** 10,015 high-quality dermoscopic images collected by Medical University of Vienna (Austria) aur Cliff Rosendahl (Queensland, Australia). (2) **ISIC 2019:** 25,331 clinical images across diverse demographics taaki model single-clinic data par overfit na ho.

Q11 HAM10000 mein Class Imbalance problem hai (Nevus 67% hai). Isko kaise handle kiya?

Answer: Class Imbalance ko humne do tariko se handle kiya: (1) **Stratified Splitting** (`stratify=meta['label']`): Isse train (70%), validation (15%), aur test (15%) har ek split mein rare classes (jaise dermatofibroma aur vascular lesions) ka exact percentage maintain rehta hai. (2) **Class Weighting** (`compute_class_weight`): Loss calculation ke dauran minority classes par higher penalty lagayi jaati hai.

Q12 Explainable AI (Grad-CAM) kya hai aur medical AI mein kyu zaroori hai?

Answer: Deep learning models ko "Black-Box" kaha jata hai kyunki yeh nahi pata chalta ki decision kyu liya. **Grad-CAM (Gradient-weighted Class Activation Mapping)** target class ke gradients last convolutional layer ke feature maps par compute karta hai aur ek 2D color heatmap generate karta hai. Isse doctor verify kar sakta hai ki model ne skin ke actual lesion boundary ko dekh kar diagnose kiya hai, na ki kisi background hair ya surgical ruler ko dekh kar!

Q13 Melanoma aur Basal Cell Carcinoma mein kya difference hai? Melanoma sabse khatarnak kyu hai?

Answer: Basal Cell Carcinoma (BCC) most common hai lekin locally grow karta hai aur rarely body mein failta (metastasize) hai. **Melanoma** skin ke melanocyte pigment cells se shuru hota hai aur bahut rapidly lymph nodes aur internal organs (brain, lungs, liver) mein spread hota hai. Iska mortality rate skin cancers mein sabse high (~75% of skin cancer deaths) hota hai agar stage 1 mein na pakda jaye.

Q14 Model ki prediction threshold kya hai agar photo clear na ho?

Answer: Hamare code mein `CONFIDENCE_THRESHOLD = 0.55` (55%)** set hai. Agar kisi blurry ya low-light photo par model ka highest probability score 55% se kam rehta hai, toh system galat assumption karne ke bajaye user ko explicitly prompt karta hai: *"Insufficient confidence for a reliable preliminary assessment"*.

Category D: System Architecture, Security & LLM Safety

Q15 Web architecture mein Node.js aur FastAPI alag-alag kyu use kiye?

Answer: Separation of Concerns: (1) Node.js/Express.js client authentication (JWT), user session, OTP generation, rate limiting, aur web requests ko ultra-fast handle karta hai. (2) Python FastAPI tensor computation, NumPy array handling, OpenCV, aur PyTorch/TensorFlow deep learning inference ko GPU/C++ speed par run karta hai. Dono REST API se communicate karte hain.

Q16 Google Gemini LLM chatbot dangerous prescription medicines prescribe na kare, iske liye kya kiya?

Answer: Humne **Clinical Safety Filter & Domain-Grounded RAG** implement kiya hai: Har AI response generation se pehle Regex aur Semantic filters run hote hain. Agar LLM oral steroids, heavy immunosuppressants, ya prescription antibiotics suggest kare, filter use instantly intercept karke OTC supportive hygiene, moisturizing, aur dermatologist physical visit se replace kar deta hai.

Q17 Kya yeh system real doctor ko replace kar sakta hai?

Answer: Bilkul nahi! DermAI ek *Clinical Decision Support System (CDSS)* aur early triage platform hai. Definitive diagnosis ke liye physical dermatoscope, biopsy, aur certified dermatologist ka histopathology report mandatory hai. Hamara system explicitly medical disclaimer ke sath patient ko qualified doctor tak navigate karta hai.

Q18 Doctor Finder feature kaise kaam karta hai?

Answer: User ke browser coordinates ko OpenStreetMap **Nominatim API** se reverse-geocode karke area identify karta hai, fir **Overpass API** se nearby clinics/dermatologists query karta hai. Network fail hone par system ke paas verified local doctors ka offline fallback database hai, jo consultation fee aur 1-click Google Maps directions provide karta hai.

Q19 Password Reset Security kaise implement ki hai?

Answer: 6-digit random cryptographically secure OTP generate hota hai jo 5-minute memory store mein expire hota hai. User ke email par verification code jata hai. Agar email SMTP offline ho, toh system examiner viva presentation ke liye safe on-screen Demo Mode me OTP auto-fill kar deta hai taaki demo kabhi fail na ho.

Q20 Future Scope mein kya add kar sakte hain?

Answer: (1) **Mobile App (React Native):** Camera frame detection ke sath taaki blurry photos na khinchein. (2) **ONNX / TensorRT Edge Quantization:** Model ko smartphone ke NPU par bina internet ke offline run karne ke liye. (3) **Hospital EHR Integration:** HL7/FHIR medical records standard ke sath connect karna.